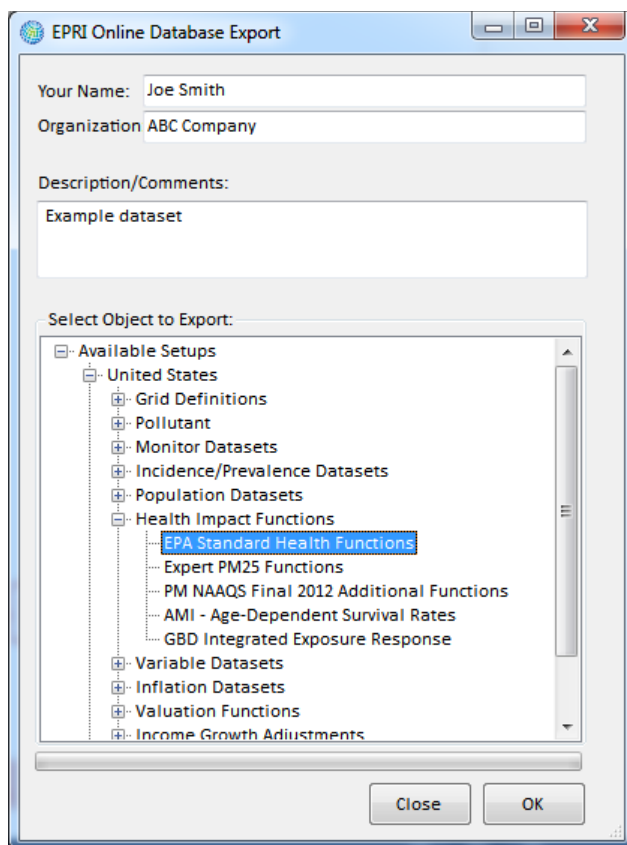
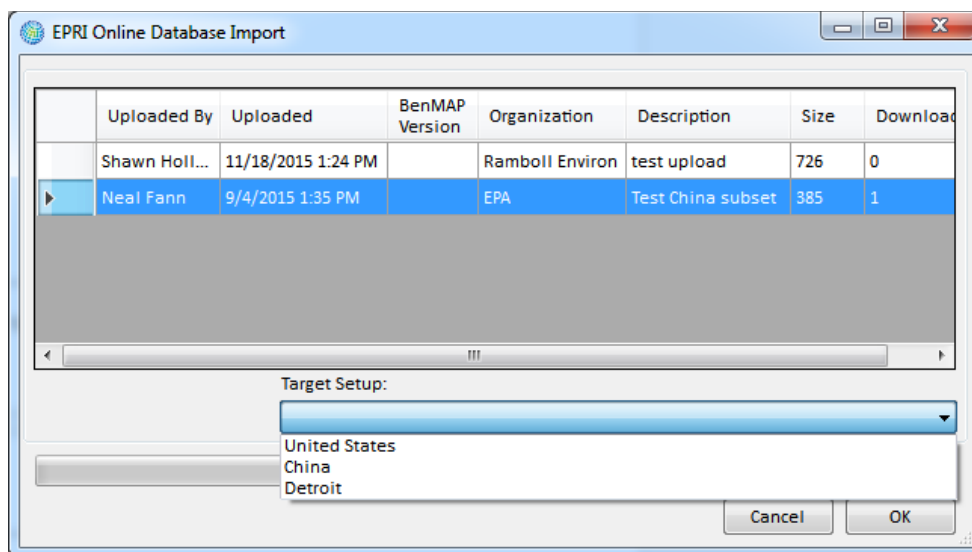




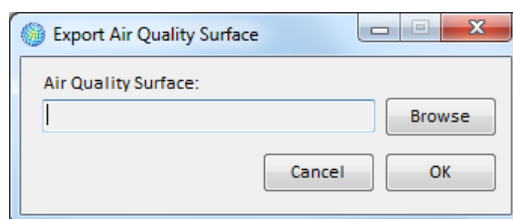
9.5 Online Database Import

The **Online Database Import** feature was designed to provide user access to an online repository of BenMAP-CE datasets shared by the user community. To import data from the online repository, select *Online Database Import* from the BenMAP-CE **Tools** menu. Select the data you wish to import by clicking on the record selector on the left side of the data grid. The record will be highlighted. Next choose your **Target Setup** from the dropdown list. Then, click the **“OK”** button. The progress bar near the bottom of the screen will update to show progress of the import. When complete, the system will notify you that the *“The database file was imported successfully.”*



9.6 Export Air Quality Surface

The **Export Air Quality Surface** tool generates a data file (.csv) reporting all of the data contained in the air quality grid. After choosing *Export Air Quality Surface* from the **Tools** menu, the **Export Air Quality Surface** window will appear. Click the **Browse** button to choose the air quality grid that you want to examine.



Click **OK** after you have selected your file. Use the **Save As** window to choose the directory where you want to save your file. And in the **File name** box, type in the name of the file.

To help keep track of what you are doing, you might want to use the same file name as your air quality grid, or something very similar. (If you use the same name, you can always distinguish the two files by the extension. An air quality grid has an .*aggx* extension and the file you are generating here has a .*csv* extension.)

When done, click the **Save** button. You can view the files you have created with any database viewer. For each **Metric** and **Seasonal Metric**, you can see the actual values. In addition, you can see the **Statistics** calculated for each. In the example below for a 12km PM_{2.5} air quality surface, you can see in the first grid cell (Column = 1, Row = 246) that the mean of the *D24HourMean* values is 9.15 and the mean of the *QuarterlyMean* values is 9.2.

Col	Row	D24HourMean	QuarterlyMean
1	246	9.15	9.2
2	246	9.16	9.2
3	246	9.16	9.21
4	246	9.17	9.21
5	246	9.17	9.22
6	246	9.54	9.65

Note that the exported files may be very large (tens to hundreds of megabytes in file size and with row counts exceeding typical spreadsheet applications). With large files, you might need to use a database program to work with the files. Alternatively, these files can also be read by simple text editors.

9.7 GBD Rollback

The World Health Organization global burden of disease (GBD) study measures burden of disease using the disability-adjusted-life-year (DALY). This time-based measure combines years of life lost due to premature mortality and years of life lost due to time lived in states of less than full health. The DALY metric was developed in the original GBD 1990 study to assess the burden of disease consistently across diseases, risk factors and regions.²⁷

The **GBD Rollback** tool uses data from the 2013 GBD study to allow users to estimate the human health burden of PM_{2.5} levels in each country as well as the benefits of reducing these air pollution levels. Users can “roll back,” or adjust ambient PM_{2.5} levels in one or more countries or regions and calculate the total burden, or avoided deaths, in that region. The tool also estimates avoided life years lost, changes in life expectancy, and the economic benefits associated with avoided deaths.

This feature is analogous to the monitor roll-back tool already available in core BenMAP-CE (discussed in Chapter 5); that tool allows users to adjust downward (or upward) air quality monitoring data in specified locations according to various algorithms (including proportional, quadratic and incremental rollbacks). The tool uses a grid with 0.1 degree resolution (approximately 10km) grid cells and includes:

- 2015 PM_{2.5} pollution concentrations²⁸. Negative concentrations were adjusted to zero.

²⁷ For more information on the GBD, see: http://www.who.int/topics/global_burden_of_disease.

²⁸ Cohen, A. J.; Brauer, M.; Burnett, R. T. (2017). Estimates and 25-year trends of the global burden of disease attributable to ambient air pollution: an analysis of data from the Global Burden of Diseases Study 2015. *Lancet*, 389(10082), 1907-18.

- 2015 global population²⁹ data stratified by age and gender. Elder populations were combined into an "80UP" age group to align with the incidence dataset.
- 2013 mortality incidence³⁰ for six health endpoints (COPD, cerebrovascular disease, ischemic heart disease, lung cancer, acute lower respiratory infection, and non-accidental) stratified by age and gender. Neonatal ("0 to 0") and "1 to 4" age groups were combined into a "0 to 4" age group to align with population data.
- The 2013 Integrated Exposure Response (IER) function³¹ employed by the 2013 GBD study to estimate premature mortality associated with ambient air pollution. The 2013 IER function estimates premature mortality from COPD, cerebrovascular disease, ischemic heart disease, and lung cancer.
- The Shape-Constrained Health Impact Function (SCHIF) developed by Burnett et al. (in preparation)³². Like the IER, the SCHIF is a meta-analytic concentration-response function developed using data from many PM_{2.5}-related epidemiological studies. Depending on the country being evaluated, the SCHIF estimates draw from mortality incidence from either (a) all non-accidental causes, or (b) a "re-attributed" incidence rate representing deaths due to COPD, cerebrovascular disease, ischemic heart disease, lung cancer, and acute lower respiratory infection).

²⁹ Center for International Earth Science Information Network - CIESIN - Columbia University. 2016. Gridded Population of the World, Version 4 (GPWv4): Administrative Unit Center Points with Population Estimates. Palisades, NY: NASA Socioeconomic Data and Applications Center (SEDAC). <http://dx.doi.org/10.7927/H4F47M2C>.

³⁰ Global Burden of Disease Study 2013. Global Burden of Disease Study 2013 (GBD 2013) Incidence, Prevalence, and Years Lived with Disability 1990-2013. Seattle, United States: Institute for Health Metrics and Evaluation (IHME), 2015. <http://ghdx.healthdata.org/record/global-burden-disease-study-2013-gbd-2013-incidence-prevalence-and-years-lived-disability>

³¹ Burnett RT, Pope CA, Ezzati M, Olives C, Lim SS, Mehta S, Shin HH, Singh G, Hubbell B, Brauer M, Anderson HR, Smith KR, Balme JR, Bruce NG, Kan H, Laden F, Pruss-Ustun A, Turner MC, Gapstur SM, Diver WR, Cohen A. (2014). An integrated risk function for estimating the global burden of disease attributable to ambient fine particulate matter exposure. *Environ Health Perspect* 122:397-403; <http://dx.doi.org/10.1289/ehp.1307049>

³² Burnett RT, Chen H, Szyszkowicz M, Fann N, Hubbell B, Pope CA, Apte JS, Brauer M, Cohen A, Weichenthal S, Coggins J, Di Q, Brunekreef B, Frostad J, Lim SS, Kan H, Pruss-Ustun AM, AARP collaborators, ACS collaborators, CTS collaborators, Canadian Breast Screening collaborators, CanCHEC/CCHS collaborators, Chinese Male Cohort Collaborators, CUELS collaborators, English Cohort collaborators, Hong Kong collaborators, NHIS collaborators, NHS collaborators, Rome Census Cohort collaborators, VHM&PP collaborators. 2017. A new approach to estimating global mortality burden from outdoor fine particle exposure. Manuscript in preparation.